

Which side of the bend are you standing on?

What did each channel contribute, and what would the next unit of spend return? Those are different questions, the second is the one budgets are set on, and a single headline ROAS cannot tell them apart.

Abstract

Objective. What did each channel contribute, and what would the next unit of spend return? Those are different questions, the second is the one budgets are set on, and a single headline ROAS cannot tell them apart. **Methods.** Check that the adapter is aliases and not a parallel type system. 7 step(s) are recorded in full below. **Results.** Everything above went through the same fit(), the same forward() and the same estimand machinery as the agriculture and clinical examples in this directory. The only marketing-specific code is the translation at the top and the three named quantities in the middle. **Conclusions.** No assumptions were recorded for this run, which is not the same as none being required.

1. Introduction

This report addresses one question: What did each channel contribute, and what would the next unit of spend return? Those are different questions, the second is the one budgets are set on, and a single headline ROAS cannot tell them apart.

2. Methods

Question. What did each channel contribute, and what would the next unit of spend return? Those are different questions, the second is the one budgets are set on, and a single headline ROAS cannot tell them apart.

Check that the adapter is aliases and not a parallel type system

This is the load-bearing claim of the whole design, so it is worth checking with `is` rather than asserting in prose. Channel IS Treatment. Geo IS Unit. KPI IS Outcome — the same objects, renamed at the edge. If these were separate classes, every function downstream would need a marketing variant, and the core would slowly acquire a domain.

Considered instead. A `MarketingTreatment` class that wraps a `Treatment`. It reads more explicitly and it is the beginning of a fork: two type hierarchies that must be kept in step by hand, and a core that gradually learns what a channel is.

```
Channel is Treatment : True
Geo is Unit : True
KPI is Outcome : True
spend('paid_search') -> name='paid_search' dimension=Dimension($)
unit='USD' description='' numeraire='USD'
```

Take the frame in the shape a planner actually hands you

Weekly rows, one column per channel, a geo and a date. Nobody in marketing holds data in axiom's internal shape, so the adapter's real job is the translation, and `role_map` makes that translation inspectable rather than buried in a reshape.

Considered instead. Asking the planner to reshape into long format first. That moves the error-prone step outside the library, where it is done differently by every analyst and reviewed by nobody.

```
role map : unit 'geo', time 'week', outcome Outcome(name='revenue',
dimension=Dimension($), unit='USD', description='', aggregation='sum')
treatments : ['paid_search', 'social']
panel : ('north', 'south', 'west') geos x 16 weeks
balanced : True
```

Let the spec default to saturation and carryover

`marketing_spec` puts a Hill kernel and a carryover on every channel by default, because `spend` does neither of the two things a linear model assumes it does: it does not return proportionally forever, and it does not stop working at midnight on Sunday. Making these the defaults rather than options means the wrong model has to be chosen deliberately.

Considered instead. A linear regression on `spend` with `adstock` bolted on afterwards. It fits, it reports a coefficient, and that coefficient is the average return over whatever spend range happened to occur — which is not the number any budget decision needs.

```
treatments : ('paid_search', 'social')
intercept : 'hierarchical'
kernels : {'paid_search': 'hill', 'social': 'hill'}
carryover : {'paid_search': 'geometric', 'social': 'geometric'}
```

Fit a world whose truth is known, so the numbers are checkable

The point of this example is not to be persuasive about marketing, it is to show that the marketing quantities come out of the same fit() as the agriculture and clinical examples. A simulated channel with a known Hill curve — its half-saturation point deliberately inside the observed spend range, so the data actually sees the bend — makes every number below gradeable.

Considered instead. Fitting the frame built above. It has no signal in it: the revenue column is zeros. A demonstration that fits noise and reports confident quantities is worse than no demonstration.

converged : True
true half-saturation spend : 30.0, inside the observed range

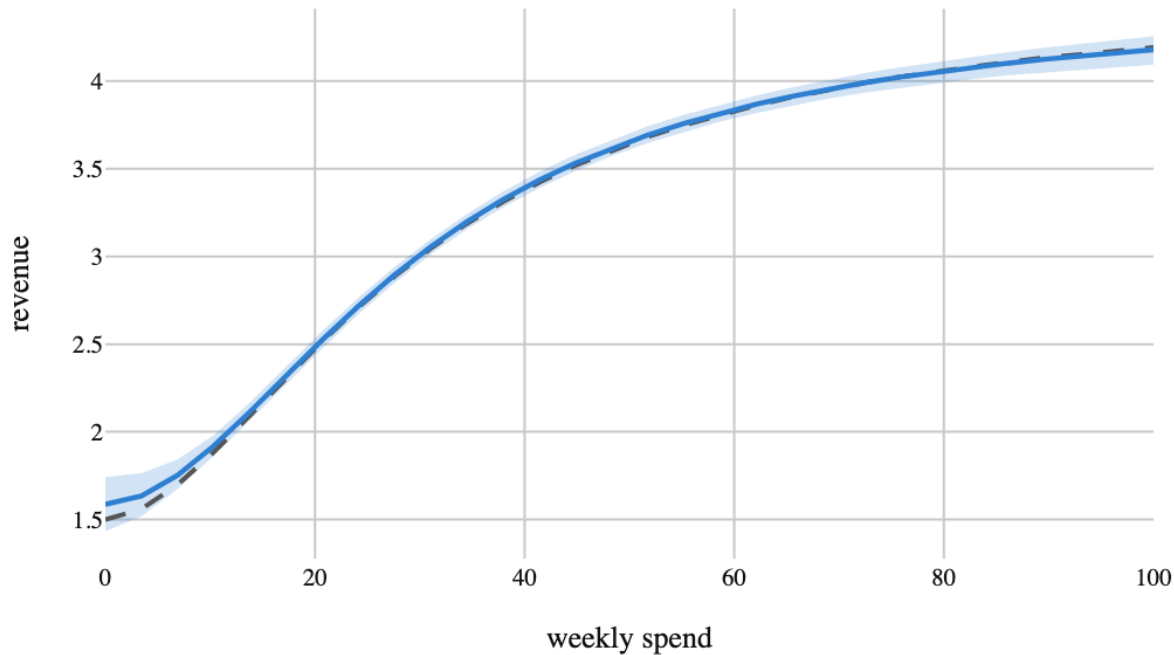


Figure 1. The response curve the fit recovered. S-shaped, not merely concave. There is a stretch at low spend where the curve is getting steeper, and everything surprising below follows from that

Ask for the three named quantities

Contribution, ROAS and marginal ROAS are three different questions and each is an estimand with its own definition, interval and provenance. Reporting them together makes the difference visible: the first is a total, the second is an average, the third is a derivative.

Considered instead. Reporting ROAS alone, which is what a marketing dashboard does. It is an average over the whole observed spend range, so it answers 'was this channel worth running' and is routinely used to answer 'should we spend more' — a question it cannot address.

```
contribution_paid_search contrast 66.2346 [58.4, 73.1] (90% HDI)
roas_paid_search ratio 0.0375 [0.0331, 0.0414] (90% HDI)
marginal_roas_paid_search marginal 0.0385 [0.0362, 0.0405] (90% HDI)
average return per unit spent : 0.0375
return on the NEXT unit spent : 0.0385
ratio : 1.02
```

estimand	kind	mean	90% interval
contribution_paid_search	contrast	66.2346	[58.4, 73.1] (90% HDI)

roas_paid_search	ratio	0.0375	[0.0331, 0.0414] (90% HDI)
marginal_roas_paid_search	marginal	0.0385	[0.0362, 0.0405] (90% HDI)

Table 1

Then look at the whole marginal curve, not one summary of it

The number that answers 'should we spend more' is the slope at the spend level you are actually at.

marginal_band gives that at every level with its uncertainty, and on a Hill curve it is not monotone: below the inflection an extra unit returns MORE than the average, above it less.

Considered instead. Comparing the single marginal ROAS to the single average ROAS and concluding 'we are past saturation'. One ratio at one implicit spend level cannot locate the bend, and the bend is the whole decision.

the marginal peaks around spend 16 and falls after.

weekly spend	return on the next unit	90% interval
4	0.0284	[0.0150, 0.0452]
8	0.0464	[0.0327, 0.0603]
16	0.0600	[0.0543, 0.0651]
24	0.0541	[0.0502, 0.0583]
32	0.0425	[0.0392, 0.0458]
40	0.0317	[0.0294, 0.0342]
55	0.0181	[0.0160, 0.0198]
75	0.0092	[0.0073, 0.0108]
100	0.0046	[0.0032, 0.0058]

Table 2

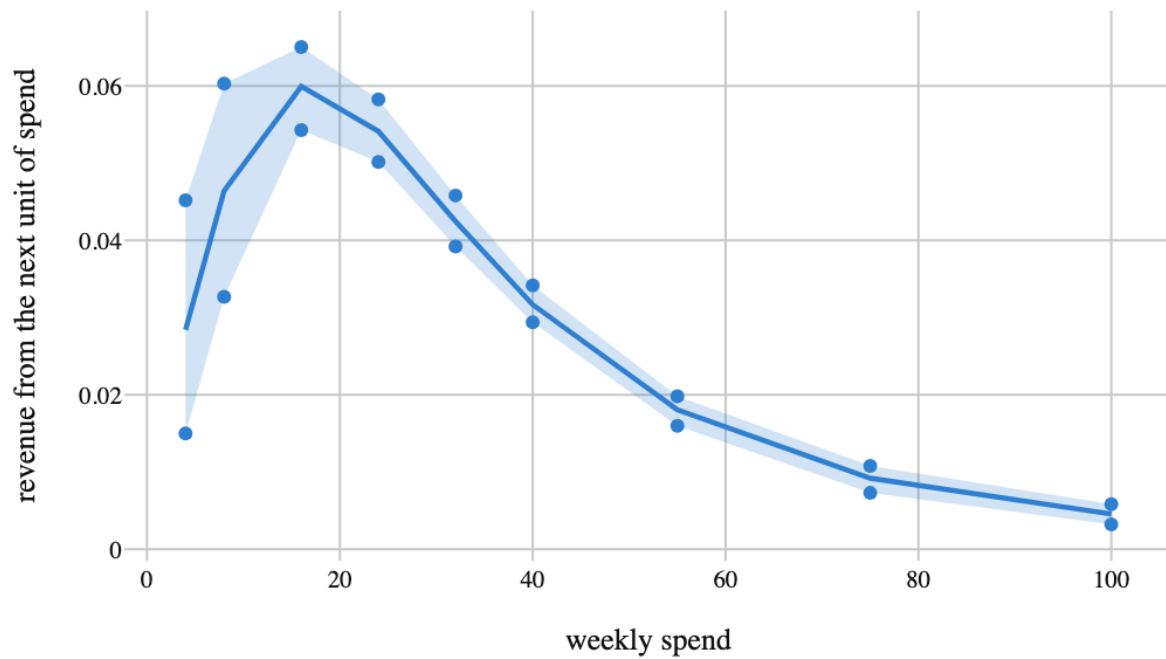


Figure 2. Return on the next unit, at every spend level. It rises, peaks near 16, and falls. Left of the peak, more spend is worth more than the last unit was. Right of it, less. A headline ROAS reports one number for this entire curve

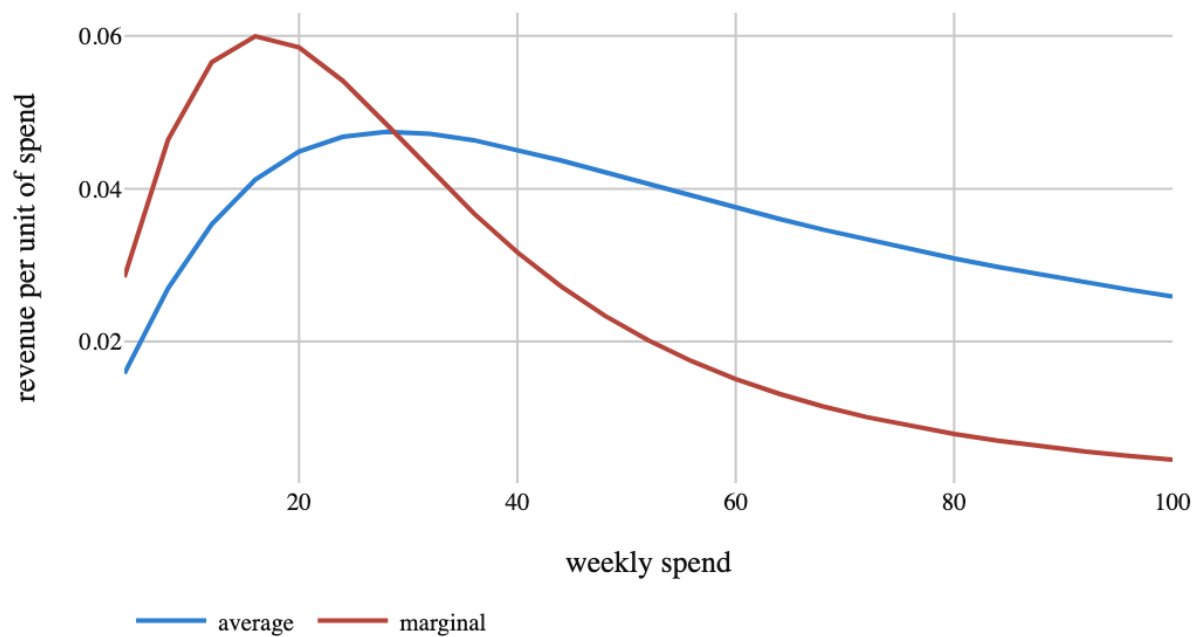


Figure 3. Average and marginal return, on the same axis. They cross. Where marginal is above average, the average is still rising and the channel is under-invested; where it falls below, every further unit drags the average down. The crossing point is the number a budget conversation is looking for, and neither summary statistic on its own can locate it

Narrow the window, because the question usually has one

'What is the marginal return' almost always means 'over the period I am planning for'. A window restricts the estimand to part of the horizon, and because it is part of the estimand rather than a filter applied afterwards, the resulting number carries which window it belongs to.

Considered instead. Slicing the data and refitting. That changes the model as well as the question, and the two effects become impossible to separate — a smaller window means less data, and less data means a different fitted curve.

marginal over the whole horizon : 0.0385

marginal over weeks 8-16 only : 0.0351

3. Results

No findings were recorded.

4. What the run showed

Written by the analyst after seeing the output, and reproduced unchanged. Nothing in this section is generated.

Everything above went through the same `fit()`, the same `forward()` and the same estimand machinery as the agriculture and clinical examples in this directory. The only marketing-specific code is the translation at the top and the three named quantities in the middle.

The headline numbers barely differ — marginal 0.0385 against average 0.0375, a ratio of 1.02 — and that near-equality is itself misleading. The marginal curve behind those summaries runs from 0.0284 at a spend of 4, up to 0.0600 at 16, and down to 0.0046 at 100. It rises before it falls, because a Hill curve is S-shaped rather than merely concave — so the next unit of spend can be worth more than the last one was.

That is a factor of 13 between the best and worst place to put the next unit, hidden inside a pair of summary statistics that agree with each other to two decimal places. 'Our ROAS is 0.038' does not say which side of the bend the budget is standing on, and it is the sentence most budgets are set with.

That is the design commitment made concrete. The mathematics is about doses and responses; the moment a domain's vocabulary leaks into it, the same code stops being usable for trials, fields, reactors and reserves — and the marketing analysis is not made any better by the leak.

5. Model checking

No diagnostics were recorded. An analysis with no model checking is not therefore sound; it is unexamined.

6. Discussion

There are no findings to interpret.

7. Conclusions

What did each channel contribute, and what would the next unit of spend return? Those are different questions, the second is the one budgets are set on, and a single headline ROAS cannot tell them apart.

No finding was recorded, so no conclusion follows.

8. Limitations

No assumption in the record is left unresolved. That is a statement about the record, not a guarantee about the world.

9. Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.